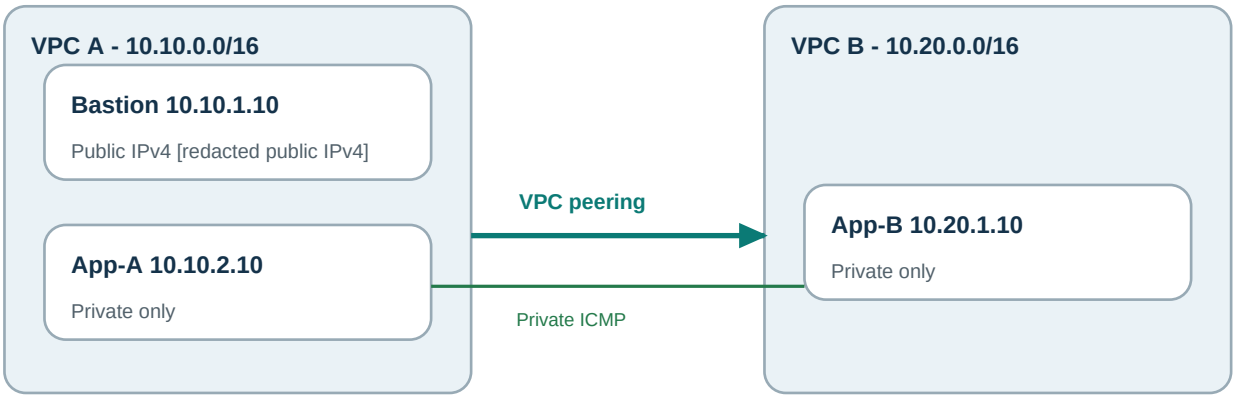
AWS VPC Peering

EWB Courses student project - deployment and verification report

Student	Tatwa Shahare
Region	Asia Pacific (Mumbai), ap-south-1
Report date	31 October 2026
Final AWS status	ewb-vpc-peering-lab - DELETE_COMPLETE

This report records the AWS deployment, route-disabled baseline, connectivity checks and cleanup in Mumbai. The baseline was captured with the peering connection present but both private inter-VPC routes temporarily removed; it demonstrates that routes are required and is not represented as a pre-creation test. The private SSH key is not included.



Laptop -> bastion -> private hosts uses SSH ProxyJump

Architecture summary

Two non-overlapping VPCs connect through one peering connection. App-A and App-B have private addresses only. A single bastion in VPC A provides SSH administration. VPC B has no Internet Gateway.

Status: Connectivity and controlled ICMP verification completed. The temporary stack was deleted after evidence capture; the standalone key pair is retained.

1. Resource inventory

Resource	Identifier / address	Status
CloudFormation stack	ewb-vpc-peering-lab	DELETE_COMPLETE
VPC A	[redacted AWS resource ID] / 10.10.0.0/16	Temporary; removed after tests
VPC B	[redacted AWS resource ID] / 10.20.0.0/16	Temporary; removed after tests
Bastion	[redacted AWS resource ID] / 10.10.1.10 / [redacted public IPv4]	SSH verified
App-A	[redacted AWS resource ID] / 10.10.2.10	SSH and ping verified
App-B	[redacted AWS resource ID] / 10.20.1.10	SSH and ping verified
VPC peering	[redacted AWS resource ID]	Active during tests
Public subnet A	[redacted AWS resource ID] / 10.10.1.0/24	Temporary; removed after tests
Private subnet A	[redacted AWS resource ID] / 10.10.2.0/24	Temporary; removed after tests
Private subnet B	[redacted AWS resource ID] / 10.20.1.0/24	Temporary; removed after tests
App-A security group	[redacted AWS resource ID]	Temporary; removed after tests
App-B security group	[redacted AWS resource ID]	Temporary; removed after tests
Bastion security group	[redacted AWS resource ID]	Temporary; removed after tests
Key pair	a separate lab key pair	Retained in AWS; private key local

The private key remains on the student computer and is not included in this package. Do not attach or publish it.

2. Routes and subnet associations

Route table	Subnet	Routes
Public A - [redacted AWS resource ID]	10.10.1.0/24	local; 0.0.0.0/0 -> IGW; 10.20.0.0/16 -> [redacted AWS resource ID]
Private A - [redacted AWS resource ID]	10.10.2.0/24	local; 10.20.0.0/16 -> [redacted AWS resource ID]
Private B - [redacted AWS resource ID]	10.20.1.0/24	local; 10.10.0.0/16 -> [redacted AWS resource ID]

3. Inbound security rules

Security group	Inbound	Source
Bastion	TCP 22	[redacted public IPv4]/32
App-A	TCP 22; ICMP Echo Request	10.10.1.10/32; 10.20.1.10/32
App-B	TCP 22; ICMP Echo Request	10.10.1.10/32; 10.10.2.10/32

No inbound SSH or ICMP was open to 0.0.0.0/0. These entries document the verified rules before stack deletion.

4. Verification results

All results below come from SSH sessions authenticated with the local key. SSH to each private host used ProxyJump through the bastion. The private destination addresses were used for peer tests.

Test	Observed result	Outcome
Bastion SSH	Connected; ip-10-10-1-10.ap-south-1.compute.internal	PASS
App-A SSH via bastion	Connected; ip-10-10-2-10.ap-south-1.compute.internal	PASS
App-B SSH via bastion	Connected; ip-10-20-1-10.ap-south-1.compute.internal	PASS
App-A -> App-B ping	4 sent, 4 received, 0% loss	PASS
App-B -> App-A ping	4 sent, 4 received, 0% loss	PASS
Route-disabled baseline ping	4 sent, 0 received, 100% loss with both private peer routes removed	EXPECTED FAILURE
Routes restored; reciprocal pings	Both directions: 4 sent, 4 received, 0% loss	PASS
Baseline limitation	The peering resource existed; both private peer routes were removed for this test.	DISCLOSED

Test summary

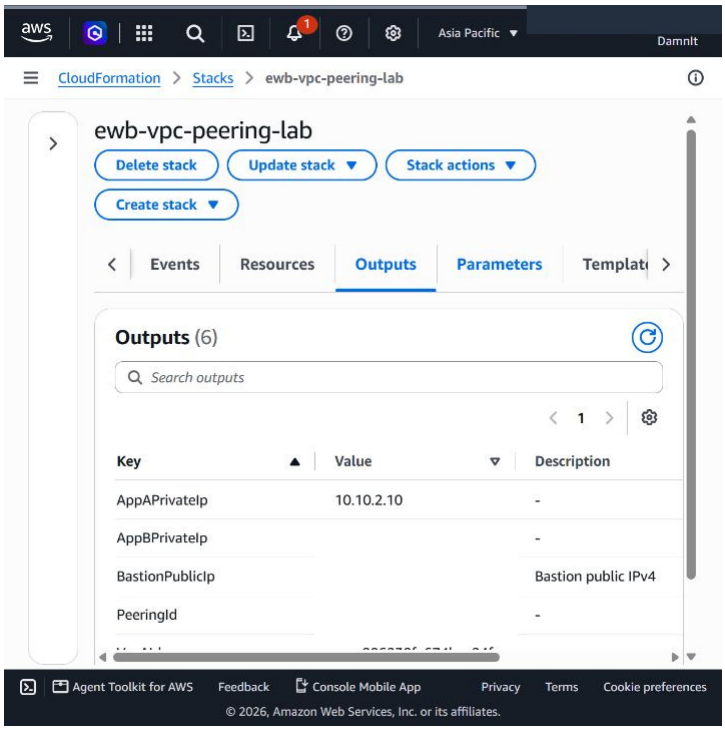
With both private peering routes removed, App-A could not reach App-B (4 packets, 0 replies). The peering resource itself existed during this baseline, so the result demonstrates the need for routes rather than a pre-creation state. After both routes were restored, App-A to App-B and App-B to App-A returned four replies. The controlled App-B ICMP test removed only Echo Request from 10.10.2.10/32 while keeping SSH from the bastion. App-A ping failed with 100% loss; after restoring the rule ([redacted AWS resource ID]), the same ping returned 4/4 replies.

Test evidence

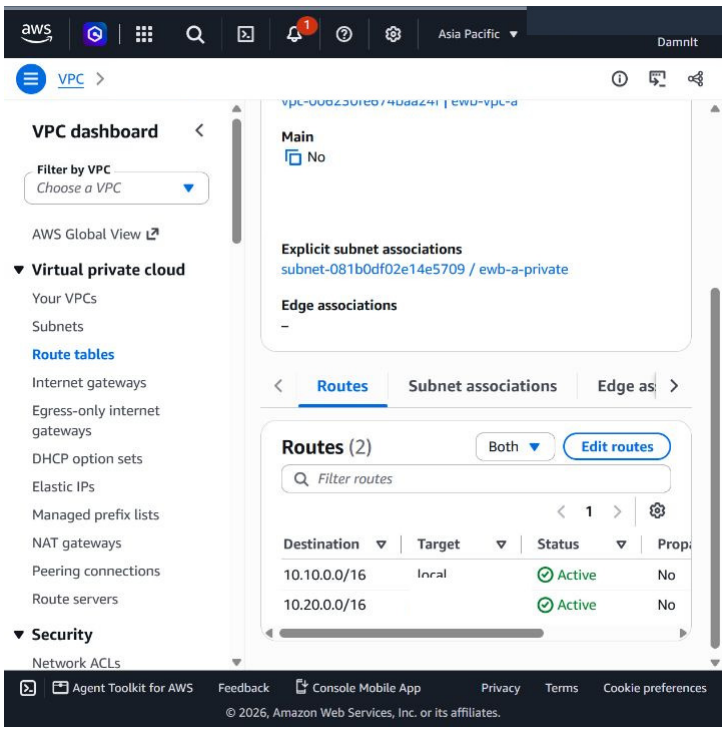
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Route-disabled App-A -> App-B: 4 packets transmitted, 0 received, 100% packet loss.
Routes restored, App-A -> App-B: 4 packets transmitted, 4 received, 0% packet loss.
Routes restored, App-B -> App-A: 4 packets transmitted, 4 received, 0% packet loss.
App-B ICMP rule removed: 4 packets transmitted, 0 received, 100% packet loss.
App-B ICMP rule restored: 4 packets transmitted, 4 received, 0% packet loss.
SSH ProxyJump reached Bastion, App-A and App-B successfully.
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5. AWS Console evidence

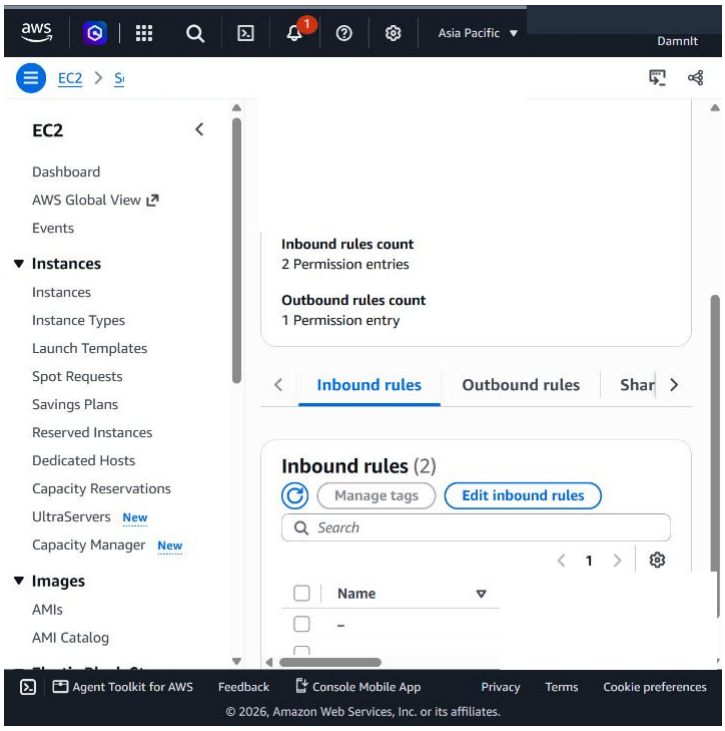
These screenshots were captured from the AWS Console during the temporary deployment and after cleanup. Terminal output is preserved as text in verification-results.txt; an original terminal screenshot was not available.



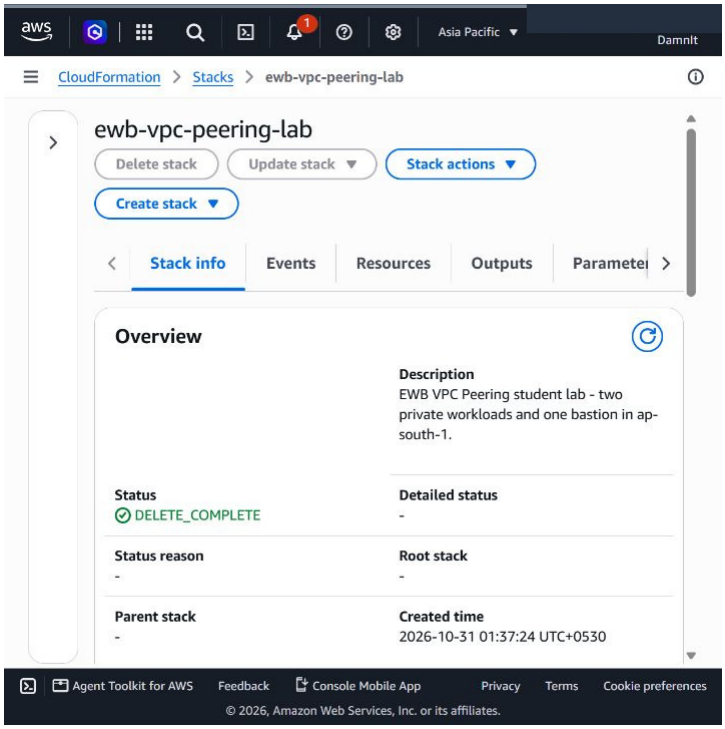
CloudFormation outputs from the live stack: private host addresses, bastion public address and peering ID.



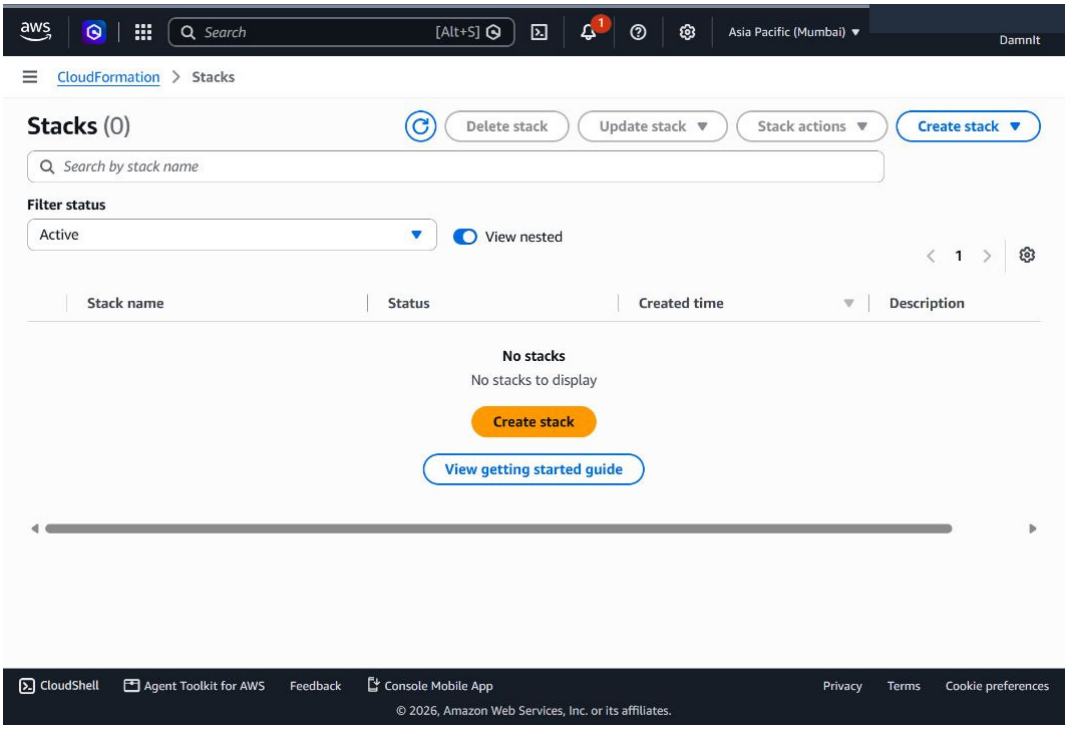
VPC A private route table showing the active route through the peering connection.



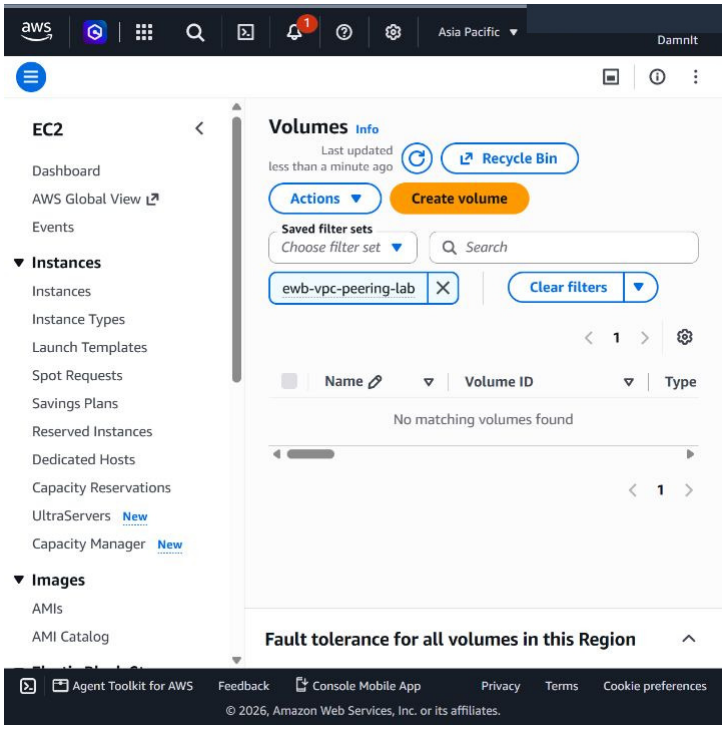
App-B security group after the ICMP Echo Request rule was restored.



CloudFormation stack status after cleanup: DELETE_COMPLETE.



CloudFormation active stack list after cleanup: no stacks to display.



EC2 Volumes search for the lab stack name: no matching volumes found.

6. Traffic path and learning

Forward path: App-A sends to 10.20.1.10. Its subnet route table sends 10.20.0.0/16 across the peering connection. App-B allows Echo Requests from App-A at 10.10.2.10/32.

Return path: VPC B has a route for 10.10.0.0/16 back through the peering connection. Security groups are stateful, so replies to allowed traffic return automatically.

Private SSH: The laptop connects to the bastion public address. ProxyJump asks the bastion to open a separate connection to App-A or App-B. The private hosts see the bastion private address, 10.10.1.10, as the SSH source.

Viva answers

Why is a return route needed? The reply must have a route back to the source host. A forward route alone does not complete communication.

Why does App-B allow SSH from 10.10.1.10/32? ProxyJump creates the second SSH connection from the bastion, whose private IP is 10.10.1.10.

Why does active peering not guarantee a successful ping? Route tables, security groups, network ACLs, host addresses, and instance state also affect reachability.

Can VPC B use VPC A Internet Gateway through peering? No. VPC peering is not transitive and does not share internet gateway access.

7. Submission and cleanup checklist

Student: Tatwa Shahare. Batch is omitted per the student request. Test summaries and cleanup checks are recorded in this report. Six AWS Console screenshots are included in this report and as separate PNG files. Terminal command output is preserved in verification-results.txt; an original terminal screenshot was not available.

The temporary stack was deleted after testing; CloudFormation shows DELETE_COMPLETE and the active stack list is empty. The EC2 Volumes search returned no matching lab volumes. Stack-managed instances, VPCs, routes and security groups were removed by CloudFormation. The standalone key pair a separate lab key pair is retained as requested. No account-wide billing or every possible resource category was checked.

Supporting files

The submission package includes the CloudFormation template, editable report, verification notes and AWS Console screenshots. The evidence recapture used an earlier AWS-stored CloudFormation template because the local upload control did not open. The included stack.yaml is the phase-enabled project source, not the exact template revision used for this recapture. The tested architecture matched; the baseline was produced by temporarily removing private routes while peering was present. Terraform files are an untested alternative and were not used. The separate GitHub-ready source package excludes personal connection details.